What we have, what would beat us, and the plan to close it

We are not starting from zero and we are not finished. This document is the honest position: an audit against the five judged metrics, the eight places a strong rival would out-argue us, and the five moves that would take this from a very good hackathon build to something that does not currently exist. **Every claim below is checked against a committed result file, not memory.**

---

01 Where we stand against the judged metrics

The weighting decides where effort belongs. Technical depth and problem fit are **55% between them**, so that is where the plan concentrates. Prototype quality is already our strongest asset and needs defending, not growing.

METRIC	WEIGHT	OUR POSITION	GRADE
Technical depth	30%		Strong
Identifiability derived, not asserted — three collinearities, two resolvable only by prior. Hand-written Kalman + RTS with exact likelihood, verified against an independent joint MVN to 1e−9. Physics validated non-circularly. **Held back by:** a compound specification error, a thermal layer that does not feed the headline number, and an evidence base of four circuits.			
Addresses the problem statement	25%		Strong, mis-aimed
The brief is *practice → clean curves → validate against race pace*. Experiment 03 is literally that. **But** our demo opens on a race, we use only FP2 and ignore FP1/FP3, and the practice→race pipeline has no dedicated product surface. We are answering the question in an appendix.			
Working prototype / feasibility	20%		Our best asset
Runs offline from one command, no Node and no API key. Nine views, 24 endpoints, live streaming replay, MCP server, 185 tests, verified from a clean clone. Most finalists will demo a notebook.			
Real-world applicability	15%		The soft flank
Adoption path and MCP integration are written and real. Missing the thing teams actually want: **a forecast before they arrive at a circuit**. No wet handling, no driver effects, no validation against real race outcomes.			
Presentation & communication	10%		Strong

36-page dossier, 24-slide deck, self-driving 4:55 demo video, README with screenshots, shareable web pitch. Needs re-cutting once the numbers below change — not rebuilding.

02 What we actually hold today

Estimator

Hierarchical linear-Gaussian state-space model in tyre age, fitted across the whole grid. Kalman filter + RTS smoother written in NumPy, six hyperparameters by L-BFGS-B on the exact likelihood.

`10,654 lines in src · 41 modules · 17,278 Python lines in all`

Evidence

Twelve recorded experiments. Ground-truth recovery `0.0044` vs naive `0.0966` over 25 sessions. Practice→race `0.0871` vs `0.1442` over **27 events and 62 comparisons**, two seasons.

`12 result files · 113 sessions · 61,396 laps · every figure read, none typed`

Product

Nine views, 3D racing line, live lap-by-lap replay at 0.22 ms/lap, hybrid retrieval over our own research corpus, MCP server with seven read-only tools.

`6,583 lines TypeScript · 24 endpoints · 185 tests`

The number that used to worry us — and what it cost to fix

This section previously read: *"our entire real-data claim rests on 10 comparisons across 5 events, all from 2024."* That is now **62 comparisons across 27 events over 2024 and 2023**, drawn from a corpus of **113 sessions and 61,396 laps** spanning three seasons.

The risk register below predicted that results might get worse at scale and said that would be *"a feature, not a failure"*. It was right on both counts. Practice→race MAE went from `0.0518` to `0.0871` s/lap: the five-event sample had been flattering us. The naive baseline worsened in step (`0.1166` → `0.1442`) and the **40% gap held**, which is the part of the claim that was ever real. Interval coverage told the same story — 90% on ten comparisons was not good calibration, it was too small a sample to detect that the true figure is 76%.

03 Where a strong rival would beat us

Ordered by how much damage each does under questioning.

01 **Compound labels hide the physics** `TESTED — HYPOTHESIS NOT SUPPORTED`

Pirelli nominates three of the C1–C5 range for each Grand Prix, so **"MEDIUM" at Monza is C4 while "MEDIUM" at Silverstone is C2 — different rubber**, and we pooled them as one compound. We predicted that was a specification error costing us accuracy. **Experiment 08 says otherwise.** Across 58 stint estimates from 23 races, pooling on the compound actually fitted transfers no better than pooling on the relative label — leave-one-event-out MAE 0.0499 against 0.0469, Wilcoxon `p = 0.16`. The likely mechanism is that Pirelli nominates to *equalise*: harder compounds for abrasive circuits, softer for gentle ones. The label already carries circuit severity, by design. The compound table is built and tested regardless, because the circuit-transfer model in M3 needs a physical compound to predict *for*.

02 **The evidence base is thin** `CLOSED — 113 SESSIONS, 61,396 LAPS`

~~Four circuits, one season, ten real comparisons.~~ The corpus now spans 2024, 2023 and 2022; the practice→race claim rests on 27 events and 62 comparisons across two seasons, and the circuit-transfer work on 140 stint estimates at 25 venues. "Would this hold over a full season?" now has an answer, and the answer is that the headline got worse and the margin held.

03 **We report the practice→race bias but never explain it**

`TESTED — MECHANISM STILL OPEN, CONSEQUENCE CLOSED`

Nine candidate explanations were tested against the signed error with a Benjamini–Hochberg correction across all nine (exp10). **Two survive:** mean practice stint length (ρ +0.37, p 0.003) and pit stops per driver (ρ +0.33, p 0.009). **Three failed:** the practice→race temperature gap, traffic, and the model's own posterior sd. The fuel-load hypothesis this card proposed was not among the survivors.

The causal story those two implied — practice going deeper into the wear curve than a pitted race — was then built and **refuted** (exp11). Practice runs are 4.4 laps *shallower* than race stints, not deeper, and forcing a common tyre-age window made bias and error both worse. The mechanism remains unidentified and is now stated as such rather than guessed at; the *consequence* is closed by conformal calibration (exp12), which widens the interval until it covers what it claims.

04 **The thermal and wear layer is unearned depth** `TESTED — EXPLORATORY, AND LABELLED SO`

The demand was: *either wire them into the estimator and show they improve a number, or present them as exploratory. Physics that changes no result is theatre.*

Tested, and it does not improve a number. exp09 now runs an ablation over feature families through the identical leave-one-circuit-out procedure, so temperature can be judged on its own rather than bundled with four geometric descriptors. Against the label-mean baseline over 140 stints at 25 venues:

```
geometry only   -7.1%   p 0.008   HURTS
thermal only    -2.5%   p 0.000   HURTS
everything      -9.9%   p 0.003   HURTS
```

Track temperature, including the squared term that lets the ridge find the bidirectional cold-graining/hot-abrasion curvature our own wear model predicts, makes out-of-sample prediction **significantly worse**. The effect is small and highly reliable, which is the worst combination to discover late: it would never have shown up as an obvious regression.

So the layer is **presented as exploratory** and no headline number depends on it. That is the second branch of the demand, taken because the first failed — not because it was easier.

---

05 **No pre-event capability** `TESTED — HYPOTHESIS NOT SUPPORTED`

Built and tested by leave-one-*circuit*-out over 140 stint estimates at 25 venues (exp09), so the model has never seen the venue in any form — exactly Thursday's situation. **Circuit geometry does not merely fail to help, it makes the label-mean baseline significantly worse** (p 0.003; p 0.001 with compound identity). At 2.4× the data this is a firm negative, not an underpowered null.

The mechanism is the one exp08 found: **Pirelli nominates to equalise** — harder rubber for abrasive venues, softer for gentle ones — so the weekend label already carries circuit severity by design, and geometry laid on top re-fits severity that is already spoken for. Pre-event prediction is still the right product goal; circuit geometry is not the route to it.

---

06 **Lap-time intervals are undercovered** `CLOSED — 95.2% ON 66,606 LAPS`

The **practice→race** half of this is fixed. Coverage was 76% against a nominal 95% — worse than the 90% previously reported, which was a ten-comparison sample failing to detect the problem. Split conformal prediction (exp12) brings it to **95%** with a distribution-free finite-sample guarantee, at the honest cost of a wider interval (±0.25 vs ±0.14 s/lap). It is shipped, not just measured: `tyremind.models.conformal`, exposed by the API as `race_projection`.

The lap-time half is now closed too, and it needed a different instrument. Split conformal requires the calibration set to be exchangeable with the test point; lap times inside a race are the opposite, because the car burns fuel, the track rubbers in and a safety car rearranges the order. A fixed quantile calibrated on the first half of a race is calibrated for a race that no longer exists.

Adaptive Conformal Inference (exp13) drops that assumption rather than hoping it holds, moving the working miss-rate by `γ(α - err)` after every lap. Measured over **66,606 held-out laps from twenty races**: Gaussian 75.5%, split conformal 89.3%, **adaptive conformal 95.2%** at a median width of 7.37 s. It lifts LightGBM from 63% to 95% at only 3.5 s of width — its predictions were always fine, its sd was wrong.

Shipped rather than measured: `AdaptiveConformal` lives in the library and the live monitor now carries an adaptive interval around its one-step-ahead forecast *together with the coverage it has actually achieved*, so the claimed 95% is auditable during the session rather than after it.

And the shape has now been inspected too (exp16), because a single coverage figure is a weak check: a model can hit 95% exactly while being far too confident in the middle and far too timid in the tails, and every number reported above would still look perfect.

The **PIT histogram** asks where the truth landed inside its own predicted distribution; under a correct distribution those values are uniform. **All six rungs come out U-shaped** — too much mass in the tails — which is overconfidence seen as a shape rather than inferred from one level. A four-race pilot had shown the state-space model hump-shaped instead; ten races showed that was a small-sample artefact, which is the argument for not stopping at the first plausible answer.

The **reliability curve** sweeps the nominal level from 50% to 99%. The Gaussian intervals sit a mean `0.191` from the diagonal; adaptive conformal sits `0.005` away — **37× closer, at every level and not only at 95%**. A method tuned to look right at one level and wrong everywhere else would fail here, and this is the only test in the project that would have caught it. The calibrator is run per race and started cold each time, so it never inherits a head start it would not have on a Sunday morning.

07 **Rain is excluded, not handled** PARTLY CLOSED — DETECTED, NOT MODELLED

Wet and intermediate compounds are still dropped at ingestion, and modelling wet degradation remains out of scope: it is a different physical process and the priors do not describe it.

What has changed is that we now notice. Dropping the wet laps and fitting whatever remained was quietly producing nonsense: Canada 2024 ran 846 of 1,272 laps on wet rubber, and the dry remainder — which sits on a *drying* track, improving faster and for longer than the saturating basis can represent — yielded **−0.151 s/lap**. A tyre getting faster with age. Precisely the impossible number we criticise the naive method for, produced by us, in silence.

`session_conditions.json` now records the wet-lap fraction of every session and every experiment refuses sessions above 40%. The threshold is **calibrated, not chosen**: Canada at 66.5% gave the impossible number, Britain at 24.4% gave +0.12 to +0.15 which is entirely ordinary, so the boundary sits between them. Rainfall alone is deliberately not a criterion — Spain and Austria both recorded rain with zero wet-tyre laps and unremarkable estimates, so rain that never forces a wet tyre is carried as an advisory rather than an exclusion.

Still open: a wet Friday is exactly when a team most needs help reading a dry race, and we still cannot offer one. Detecting that we should keep quiet is a floor, not a product.

---

08 **No driver effect** TESTED — RELIABLE BUT NOT USEFUL

Tested across **1,055 driver-races, 26 drivers, 56 races** (exp15), and the result is the most interesting negative in the project because it splits into two halves that disagree.

Reliable. Split the corpus into random halves and the ranking of drivers reproduces: split-half r = **+0.36**, with a 5th percentile of +0.09, so the correlation stays positive across 500 random partitions. Some drivers genuinely are harder on tyres than others, measurably so.

Not useful. Knowing that does not help. Predicting a driver's deviation at a held-out race from their own history scores 0.0251 s/lap against **0.0248 for simply predicting the field mean** — slightly worse, p = 0.27. The reason is the ratio: the spread of driver means is 0.0110 s/lap while the race-to-race scatter around them is 0.0248. The trait is real and about **half the size of the noise it must be read through**, so on any single race a driver's history is worth less than the average.

And it may not be the driver at all. Driver and car are perfectly confounded. Teammates share a car, so the within-team difference removes it — and that contrast is *not* stable (5–95%: −0.14 to +0.70). Even the reliable part cannot be attributed to the driver rather than the machinery.

A per-driver term would add a parameter, a maintenance burden and a persuasive story, and would not improve a forecast. **Not built, deliberately.**

---

04 The five moves that decide this

Each states what it buys, which metric it moves, and — the part that matters — **the result that would prove it wrong**. A plan without falsifiable tests is a wish list.

M1 · Season-scale evidence

TECHNICAL DEPTH · PROBLEM FIT

Pull the full 2024 season — 24 events × FP1/FP2/FP3/Q/R — then 2023 and 2022. Re-run every experiment at scale. Ten practice→race comparisons become several hundred; four circuits become twenty-four. This is the foundation the other four moves stand on, and it is mostly download time rather than research risk.

> **Falsified if:** error grows or interval coverage falls when the sample grows. That would mean our current numbers are a small-sample artefact — which we would then have to report.

M2 · Model the compound that was actually fitted

COMPLETE · NEGATIVE RESULT

Done, and the hypothesis did not survive. The per-event C1–C5 allocation table is built, cited row by row and covered by tests. But identity pooling does not beat label pooling on held-out events (0.0499 against 0.0469 s/lap, `p = 0.16`), and the mechanism is interesting on its own: Pirelli nominates to equalise, so the relative label is already a circuit-adjusted compound index. The table stays because M3 needs a physical compound to predict for — but the claim is now a published negative result, not a correction.

> **Falsified if:** identity pooling does not beat label pooling on held-out events. **Outcome:** exactly that. Reported rather than retried until it agreed with us.

M3 · Predict a circuit we have never run

APPLICABILITY · TECHNICAL DEPTH

Regress compound degradation on circuit characteristics we already compute from GPS — per-corner frictional energy, lateral load spectrum, corner count, the share of the lap spent above a load threshold — and predict the degradation rate for a compound at a circuit absent from training. **This is the product.** It is the difference between explaining Friday and being useful on Tuesday.

> **Falsified if:** leave-one-circuit-out prediction is no better than the global compound mean. That is the baseline to beat, and it is not a soft one.

M4 · Explain the practice→race bias, or characterise it properly

PROBLEM FIT · TECHNICAL DEPTH

Test the fuel-load hypothesis across sessions rather than within a stint, where we already proved it cannot work (per-lap energy varies only 2.3% inside a stint). If a fuel-weighted degradation clock removes the offset out-of-sample, the brief's central validation question goes from "honestly reported caveat" to "solved". If it does not, we will know why, on a season of data instead of five events.

> **Falsified if:** the bias survives the correction on held-out events, or flips sign across the season — which would mean it is not a fuel effect at all.

M5 · Calibration rigour

TECHNICAL DEPTH

Replace coverage counting with proper probabilistic diagnostics: PIT histograms, reliability diagrams, sharpness against calibration. Diagnose the 73% lap-time undercoverage — most likely a heavy-tailed observation model fitted with Gaussian assumptions, which is testable. A model that reports intervals must be able to defend them.

> **Falsified if:** the PIT histogram is uniform and the undercoverage turns out to be a scoring artefact rather than a model defect. That is a fine outcome — it just needs proving.

05 Sequence

Ordered by dependency, not preference. Nothing in Phase 2 can start before the data in Phase 0 exists. The durations assume focused work and will move; the *order* should not.

PHASE 0 Foundation

~1 day, mostly unattended

- **Season scrape.** All 24 events of 2024, every session type, cached as Parquet. Then 2023 and 2022. Runs unattended; start it first so it downloads while other work proceeds.
- **Compound allocation table.** Per-event C1–C5 nomination, committed as data with its source cited, plus a test that every event in the cache resolves.
- **Quality audit at scale.** Our exclusion rules were tuned on four circuits. Re-check the discard rate across twenty-four — a rule that drops 68% at Monza may drop 95% at Monaco.

PHASE 1 Re-establish every claim at scale

~2 days

— **Re-run experiments 01–07** on the full corpus. Publish the deltas honestly, including any that get worse.

— **Compound-identity re-specification (M2)** and the head-to-head against label pooling.

— **Full C-MAPSS test set.** Batch the fit per engine so all 100 engines score, removing the "indicative, not like-for-like" caveat.

— **Fix the practice emphasis.** The brief is about practice sessions; make FP1/FP2/FP3 first-class and lead the demo with them.

PHASE 2 The two new capabilities

~3 days · highest risk, highest return

— **Circuit transfer model (M3)** with leave-one-circuit-out validation, and a product surface that answers "what will C3 do here on Sunday" before anyone has run.

— **Practice→race bias resolution (M4)**, tested out-of-sample across the season.

— **Driver random effect.** Does attributing part of the stint deviation to the driver improve recovery? Cheap to add, and either answer is publishable.

PHASE 3 Rigour and honesty

~2 days

— **Calibration suite (M5)** — PIT, reliability, sharpness — surfaced in the product, not just the appendix.

— **Wet regime.** At minimum a detector that says "this model does not apply and here is why". Better: a wet degradation model, which nobody expects and everybody will notice.

— **Thermal layer decision.** Wire it into the estimator and prove it improves a number, or relabel it exploratory. No middle position.

PHASE 4 Product and story

~2 days

— **Practice→race workflow view** — the literal deliverable of the brief, as a first-class screen rather than a section inside "Does it work".

— **Cost of a wrong call**, quantified against real 2024 race outcomes rather than illustrative arithmetic.

— **Re-cut everything** — dossier, deck, video, README — against the new numbers. The self-driving demo regenerates from one command, so this is cheaper than it sounds.

06 If there is less time than the plan assumes

The order that maximises judged score per hour, if we have to stop early at any point.

#	DO THIS	METRIC WEIGHT	WHY IT IS FIRST
1	**Season scrape**	30 + 25	Unattended, and every other claim gets stronger for free. Start it before anything else.
2	**Compound identity**	30	Fixes a genuine specification error. Defensible as a contribution on its own.
3	**Practice-first reframing**	25	Cheapest large gain in the whole plan. We already do what the brief asks; we present it as an appendix.
4	**Circuit transfer**	15 + 30	The business case. Risky, so it needs a real time box — but nothing else differentiates us this much.
5	**Calibration suite**	30	Turns our weakest technical answer into a strength, and it is bounded work.
6	**Re-cut the story**	10	Last, because it depends on final numbers, and our presentation is already strong.

07 What could go wrong, and what we do about it

The season scrape is slow or rate-limited

FastF1 caches per session and telemetry is heavy. **Mitigation:** start it first, run it unattended, and prioritise race + FP2 for all 24 events before adding FP1/FP3 and prior seasons. Partial coverage is still a large improvement on four circuits.

Results get worse at scale

Entirely possible — four circuits may have flattered us. **Mitigation:** this is a feature, not a failure. Report the honest revised number and the reason. A model whose error is stable across 24 circuits at a slightly worse level is more credible than one that looks perfect on four.

Circuit transfer does not beat the compound mean

The highest-risk item. **Mitigation:** time-box it, and publish the negative result with the mechanism if it fails — as we did with the energy clock. A characterised failure is worth more than a silent retreat, and we already have a track record of doing exactly that.

Scope creep eats the presentation

The plan is deliberately larger than the time. **Mitigation:** the priority ladder in section 06 is the contract. Everything below the line we reach is documented as future work with its falsifiable test attached, which reads as rigour rather than omission.

08 What "finished" looks like

Written now, so we are not deciding it under time pressure later.

METRIC	THE SENTENCE WE WANT TO BE ABLE TO SAY, AND DEFEND
Technical depth 30%	"We derived why this problem is unidentifiable, resolved what can be resolved, corrected a compound specification error nobody else models, and validated on a full season with proper probabilistic calibration — publishing the results that went against us."
Problem fit 25%	"From practice alone we produce a clean degradation curve per compound, and we validate it against race-day pace across a full season — which is precisely what the brief asked for, as the first screen of the product."
Prototype 20%	"One command, no network, no key. It runs live at 0.22 ms a lap and an agent can call it. Here is a clean clone doing it."
Applicability 15%	"We can tell a team what the tyres will do at a circuit before they arrive, and we can show the leave-one-circuit-out error on that claim."
Presentation 10%	"Every number on every slide is read from a committed file. Regenerate the experiments and the claims regenerate with them — including the ones that do not flatter us."

The one-sentence version, for the room

Everyone in this final will show a model that predicts tyre degradation. **We are the only ones who can show whether the reason our model gives is the right one** — and, by the end of this plan, the only ones who can predict a circuit we have never run.